Diffraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain spreading around edges and through openings.

SUCCESS CRITERIA - I can...

1. Compare opening size with wavelength.
2. Distinguish diffraction from refraction.
3. Use a wave picture to justify spreading.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A wave travels **9 m** in **0.3 s** at frequency **50 Hz**. **Find speed** and wavelength.

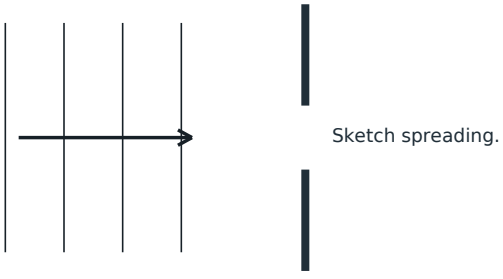
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: λ = v/f; **compare opening width** with λ. Spreading is pronounced when the opening width is comparable to wavelength. **Label the opening** and wavefronts.

Worked reasoning: A wave with wavelength **1 m** meets openings **1 m** and **10 m** wide. Spreading relative to opening size is more noticeable for the **1 m** opening.

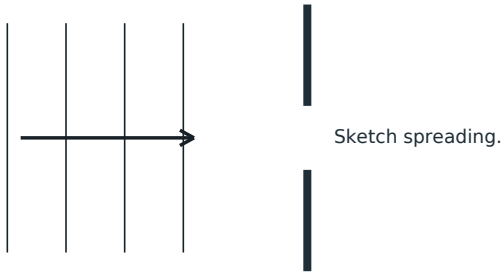
Your turn: Draw incoming parallel wavefronts and the emerging patterns for each opening.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch wavefronts after a narrow** opening about one wavelength wide.
2. Compare with a much wider opening.

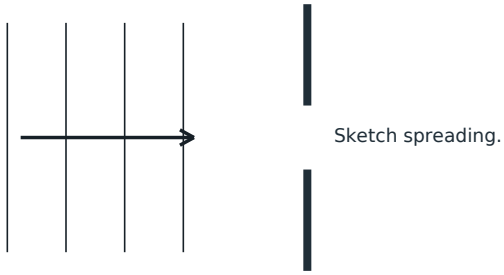


Marzano check - circle: ① ② ③

04 Apply the physics

A **340 Hz** sound travels at **340 m/s** toward a doorway **0.8 m** wide.

1. **Calculate its wavelength** and **compare wavelength**/opening ratios for this sound and visible light of wavelength **5.0×10^{-7} m**.
2. Use wavefront sketches to **explain why hearing around a** doorway differs from seeing around it.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

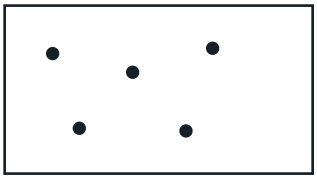
A student calls all wave bending “refraction.” Explain how diffraction can happen without crossing into a different-speed medium.

Marzano check - circle: 1 2 3

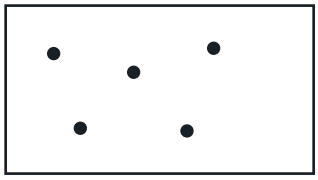
06 Review: connect an earlier idea

For a fixed ideal-gas particle mass, average translational kinetic energy becomes **16** times as large. **Compare rms speed**.
Use particle arrows and labels to **show the comparison**.

Sample A



Sample B

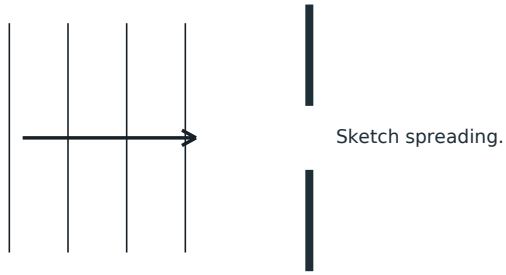


Add velocity arrows; use lengths to show a range of speeds.

Marzano check - circle: 1 2 3

07 Exit check

At fixed wave speed, frequency doubles. **Explain how** λ and typical diffraction through a fixed opening change.

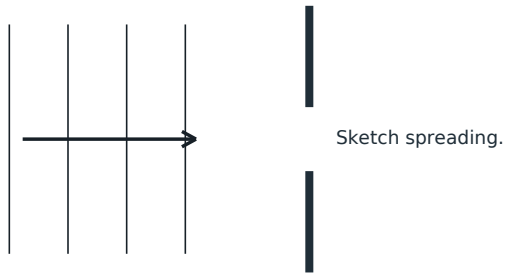


Sketch spreading.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Sound speed is **340 m/s**.
1. **Compare wavelengths** at **170** and **1700 Hz** with a **1 m** doorway.
2. **Explain which should diffract more** noticeably.



Sketch spreading.

Marzano check - circle: 1 2 3